

“Choosing The **RIGHT MATERIAL** Is **CRITICAL** To The **SUCCESS** Of The **3D PRINT**”

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3D printing has revolutionised the manufacturing sector through local and customised production. It remains one of the biggest technological trends. Benjamin Tan, vice president, Ultimaker Asia Pacific, discusses with Ayushee Sharma the various steps involved in getting the perfect 3D-printed product for an application while keeping in mind the environmental impact of such a process

Q. What are the application areas of 3D printing?

A. In the professional/enterprise market, designers, engineers, manufacturers, architects and medical specialists use 3D printers to print functional prototypes, manufacturing tools, production parts like jigs and fixtures, as well as architectural/medical models that require full geometrical freedom capabilities, industrial-grade material properties, repeatability, high uptime and an integrated workflow.

Researchers, lecturers and students in universities, tertiary institutions and institutes of higher learning use 3D printers to print functional prototypes, customised tools, and final parts that require customisation, repeatability and full geometrical freedom capabilities.

Q. How do you choose the right material to ensure print success?

A. Choosing the right material is critical to the success of the 3D print. Deciding on a material requires consideration on a few fronts. This includes design of the object—whether it needs to be flexible or extra durable, whether it is a complex shape that needs additional support to print and so on. Once these issues are sorted out, it is relatively straightforward to decide on the ideal material(s).

Q. What are the common steps involved in 3D printing?

A. To produce a 3D object, it is necessary to first have a virtual design using computer aided design (CAD) software. The CAD file needs to be converted into Standard Tessellation Language, a commonly-used language. Users have to determine and set orienta-

tion and size of the object to be printed.

After this, settings should be configured in the software, and the file should be sliced. When the digital file is ready, it is time to prepare the 3D printer. This means loading the materials and print cores that are necessary to perform the print operation.

Duration of printing varies based on complexity and size of the part. Finally, the printed part should be carefully removed and sanded, if necessary.

Q. What are the advantages of 3D-printed products?

A. There are many advantages to 3D printing and products but more commonly reported benefits from businesses include faster production, cost-effectiveness, better quality, creative designs and customisation options, unlimited shapes and geometry.

3D printing is faster than conventional manufacturing methods such as injection moulding and polymer casting, thus significantly improving productivity. A printed prototype allows close examination on whether modifications are needed before embarking on mass-printing. This helps save costs before committing to bulk print orders.

3D printing also allows step-by-step assembly of the part to ensure enhanced designs and better printouts.

Traditional manufacturing techniques, though good for mass production, result in a standard template of designs. 3D-printed parts can be easily customised for uniqueness and/or personalisation. 3D printing allows printing of geometrically-complex shapes as long as proper support

material is given. This contrasts with traditional methods like moulds and cutting technologies, which can be costly.

Q. How can reliability and safety be ensured?

A. 3D printers can be designed and built for fused filament fabrication for various high-quality plastics like PLA, ABS and CPE within a commercial/business environment. Efforts must be made to match material properties with machine settings.

The mixture of precision and speed makes Ultimaker 3D printer great for concept models, functional prototypes and production of small series.

Q. What measures can be adopted to mitigate the health and environmental impact of 3D printing?

A. An important factor is selection of print material. Material guides offer data and instructions that take into consideration technical properties, aesthetic qualities and processing ability of the printed part. This is why polylactic acid, a biodegradable polymer, is popular for prototyping 3D models. It is a reliable and easy-to-print material that can be printed at low temperatures.

A fully-enclosed build chamber creates an inside-out airflow, and EPA filter removes ultra-fine particles. For example, Ultimaker S5 is integrated with Air Manager to ensure the right chamber temperature with a completely controlled air suction and ultra-fine particle filtering. These features are as important for print quality as it is for a safer work environment, especially when printing with an extended range of materials. **EFY**